LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	35
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Principle of operation of gyro magnetic compass system.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Principle of operation of gyro magnetic compass system.
5. Specific Objectives	15.1.12.4 Trainee will be able to explain gyro magnetic compass system.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of Gyro Magnetic Compass alignment procedure; Safety: Ensure electrical isolation before handling; Supervision: Instructor monitors alignment steps.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the two fundamental properties of a gyroscope?
(b) Why is a magnetic compass unreliable near the poles?
(c) How does the gyro magnetic compass differ from a standard magnetic compass?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Gyroscopic properties in navigation • Need for gyro magnetic compass • Sensing element and torque motor • Control system and damping mechanism • Alignment with magnetic north	L & Ds	PPT, IB	15 min
(b)	• Error correction mechanisms • Latitude error compensation • Speed error compensation • Automatic alignment process • System monitoring indicators	L & Ds	PPT, PDF	20 min
(c)	• Operational limitations • Power failure effects • Maintenance requirements • Troubleshooting common faults • Integration with flight instruments	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the gyro magnetic compass corrects for latitude error.
(b) Describe the role of the torque motor in the system.
(c) What happens to the compass indication during a power failure?

12. Home Assignment and Follow Up	Prepare a brief note on the differences between the gyro magnetic compass and the directional gyro, focusing on their respective roles in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Flight Lieutenant, P No. 12345	Senior Instructor, Wing Commander, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min